



A bibliometric and content analysis of financial statement fraud: focus on the use of Industry 4.0 technologies

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ABSTRACT

The main concern associated with financial statement fraud (hereinafter FSF) is its ability to compromise the integrity of the financial system, erode investor confidence, and negatively impact economic outcomes. The objective of this research is to present a global overview of the evolution in the field, as well as to describe current and emerging trends in FSF research during the period spanning the years 2000 to 2025. To achieve this, a bibliometric analysis is conducted on a database of 823 research articles collected from the Scopus and Web of Science databases. Additionally, a content analysis of the most cited articles is performed. The results indicate the existence of nine research clusters. From a financial, economic, and accounting perspective, the article highlights that fraud is an endogenous phenomenon, driven by economic incentives, financial pressures, and governance weaknesses, rather than technological limitations. The conclusions indicate that its prevalence and types vary depending on the macroeconomic and sectoral context, intensifying in crisis scenarios. New technologies should be considered as complementary tools that enhance transparency, but their impact depends on their alignment with regulatory frameworks, internal controls, and business practices.

1. Introduction

The term corporate fraud is used to describe actions taken by managers that benefit certain stakeholders at the expense of others, rather than acting in the interest of the company (ACFE, 2004). According to the Association of Certified Fraud Examiners (ACFE), the main forms of corporate fraud are corruption, asset misappropriation, and the preparation of false financial statements; the latter represents the most costly form of fraud, with potential losses reaching 5% of a company's annual revenue (Shahana et al., 2023).

Financial statements (FS) are reports that include balance sheets, income statements, and cash flow statements, reflecting the financial situation of organizations, making them essential for decision-making by investors, lenders, and other stakeholders (Ashtiani & Raahemi, 2022; Hajek & Henriques, 2017), reports that must comply with International Financial Reporting Standards (IFRS) to assess performance and operational viability (IFRS, 2021), even more so when they may contain anomalies (fraud/error) that disguise the financial reality of the organization (Chen et al., 2024).

Indeed, financial statement fraud (FSF) encompasses a range of deceptive practices that conceal the actual financial information.

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According to Blanco et al. (2023) y Parlindungan et al. (2017) the most common fraudulent practices include revenue manipulation, which creates a better appearance of profitability; expense manipulation, which involves overestimating or underestimating costs to affect economic benefits, as well as manipulation of assets, liabilities, and equity. Other types of FSF are the misrepresentation of transactions, where fictitious operations are created to inflate financial results, and the omission of significant event information that affects users' decisions (Parlindungan et al., 2017); frauds that compromise the integrity of financial information and undermine the trust of investors, regulators, and other stakeholders in the company.

The literature examines FSF from various perspectives, studies include analyses of financial pressure and related-party transactions that contribute to fraud (Hudayati et al., 2022). Others investigate the influence of financial health and management practices on vulnerability to fraud (Hasan et al., 2018) and the reliability of the Beneish M-Score model in detecting fraud in public companies (Mohamad et al., 2016). Some literature reviews (RSL) focus on the use of forensic accounting techniques (Mustafa et al., 2024) or the relationship between creative accounting and FSF (Gupta & Kumar, 2020). Other RSLs analyze the growing interest in the use of Industry 4.0 technologies (e.g., AI, machine learning, data mining) in the prevention and detection of financial fraud (Al-Hashedi & Magalingam, 2021; Ashtiani & Raahemi, 2022; Rao & Mandhala, 2024; Raudha & Saeedi, 2019).

Despite the advances in understanding FSF represented by previous research, it is essential to conduct a bibliometric review of FSF due to the lack of a comprehensive analysis that synthesizes theoretical and methodological advancements in this field (Ramírez-Alpízar et al., 2020). Despite the existence of RSL, few employ a bibliometric approach that tracks research progress and emerging trends (Ramírez-Alpízar et al., 2020; Shahana et al., 2023); a key approach to identifying gaps, trending topics, and guiding future research, as well as being a tool for entrepreneurs to learn about existing practices and strategies to minimize this scourge internationally.

The objective of the review is to provide a comprehensive understanding of trends and developments in the field of FSF. To achieve this, this research focuses on three objectives:

RO1: Identify seminal articles, authors, countries, institutions, journals, and trends in the knowledge domain of FSF.

RO2: Analyze the knowledge structure underlying the FSF research field and its evolution in terms of interrelated theoretical concepts.

RO3: Examine research trends addressing detection methods, crisis impact, and emerging technologies in accounting manipulation and transparency.

RO4: Identify the role and centrality of Industry 4.0 technologies in FSF research.

RQ5: Explore emerging trends and future directions in research on the use of Industry 4.0 technologies for the detection and prevention of FSF.

This study primarily contributes to the disciplines of accounting, finance, and economics by clarifying the intellectual foundations of research on FSF. Theoretically, it advances forensic accounting by providing recognition and precise delineation of fundamental articles, relevant authors, leading countries and institutions, and the main journals influencing the academic debate on accounting manipulation and financial transparency (Al-Hashedi & Magalingam, 2021; Ashtiani & Raahemi, 2022; Rao & Mandhala, 2024; Raudha & Saeedi, 2019). The article further explains the definition of fraud as an economic behavior, illustrated by specific accounting practices and indicators, while simultaneously presenting an integrated definition of the fundamental constructs underpinning FSF research.

Methodologically, the study provides significant implications for accounting professionals by demonstrating how bibliometric tools are applied to a field of knowledge in the accounting domain. Approaches such as co-occurrence analysis, co-authorship networks, and scientific mapping support the complex analysis of topics related to theory, detection strategies based on accounting variables, and the development of analytical models for minimizing accounting fraud. These tools assist researchers in economics and finance in systematically understanding how traditional financial analysis is gradually integrating Industry 4.0 technologies such as AI and machine learning (Nigro, 2025; Xiuguo & Shengyong, 2022). The influence of the study extends beyond the technical realm, encompassing issues of utmost importance for business management and financial regulation.

The results also facilitate, at a social and professional level, the discovery of research topics that suggest a higher possibility of FSF, including benefits management, audit risk analysis, weak corporate governance, and limitations in accounting disclosure; fundamental aspects for auditors, regulators, and analysts responsible for protecting the integrity of SF. Similarly, the results provide empirical grounding to assist in the implementation of policies that promote transparency and prevent errors and fraud, as well as to protect the interests of users of accounting information (Ramírez-Alpízar et al., 2020; Shahana et al., 2023).

Furthermore, the study highlights the importance of Industry 4.0 technologies in FSF research and provides accounting and finance scholars with a deeper understanding of how emerging digital tools are changing the way accounting manipulation is detected and managed (An & Suh, 2020; Ravisankar et al., 2011). The article examines, rather than specifically evaluating the performance of algorithms, how these technologies bridge traditional transparency rules and controls, allowing the business academic community to see where and how artificial intelligence complements existing audit policies and financial analysis methods.

Finally, the study positions bibliometric analysis as a methodological transition between technological innovations and accounting studies. The evidence presented provides information for auditors, regulators, and risk managers to update their practices, adopt new technologies, and predict behavioral patterns that could undermine the integrity of the SF (Ashtiani & Raahemi, 2022; Hajek & Henriques, 2017). The article offers a guide for researchers studying accounting, finance, and economics, as well as other technical topics, incorporating new lines of applied research on the use of Industry 4.0 technologies to enhance the transparency and quality of financial information and fraud prevention through transparency in financial processes, which strengthens professional approaches in

financial oversight, financial management, and reliability in corporate reporting.

The structure of the article is as follows: first the introduction is presented, followed by the methodology (data collection process) and then the results of the main thematic groups identified are presented. Finally, the final reflections covering the conclusions and limitations of the study are provided.

2. Methodology

To address the research questions posed in this study, a performance analysis and scientific mapping procedure was conducted using a bibliometric method. As illustrated in Fig. 1, the methodological process begins with data collection and culminates in the creation of the strategic diagram.

Below is the methodology organized into the following subtopics.

2.1. Data collection and processing

Fig. 1a shows the equation and search process for data collection, where the Web of Science and Scopus databases were used, covering various disciplines and international academic journals. An initial search was conducted using the terms "finan* statements" and "fraud," with the search including the title, abstract, author, and keywords of the articles. A total of 823 records corresponding to the period 1979 and 2025 were obtained as a result. Subsequently, an initial analysis was performed based on the type of document (scientific and review articles), and a cleaning process was carried out to identify and exclude duplicate articles; this process was based on a preliminary review of titles and abstracts.

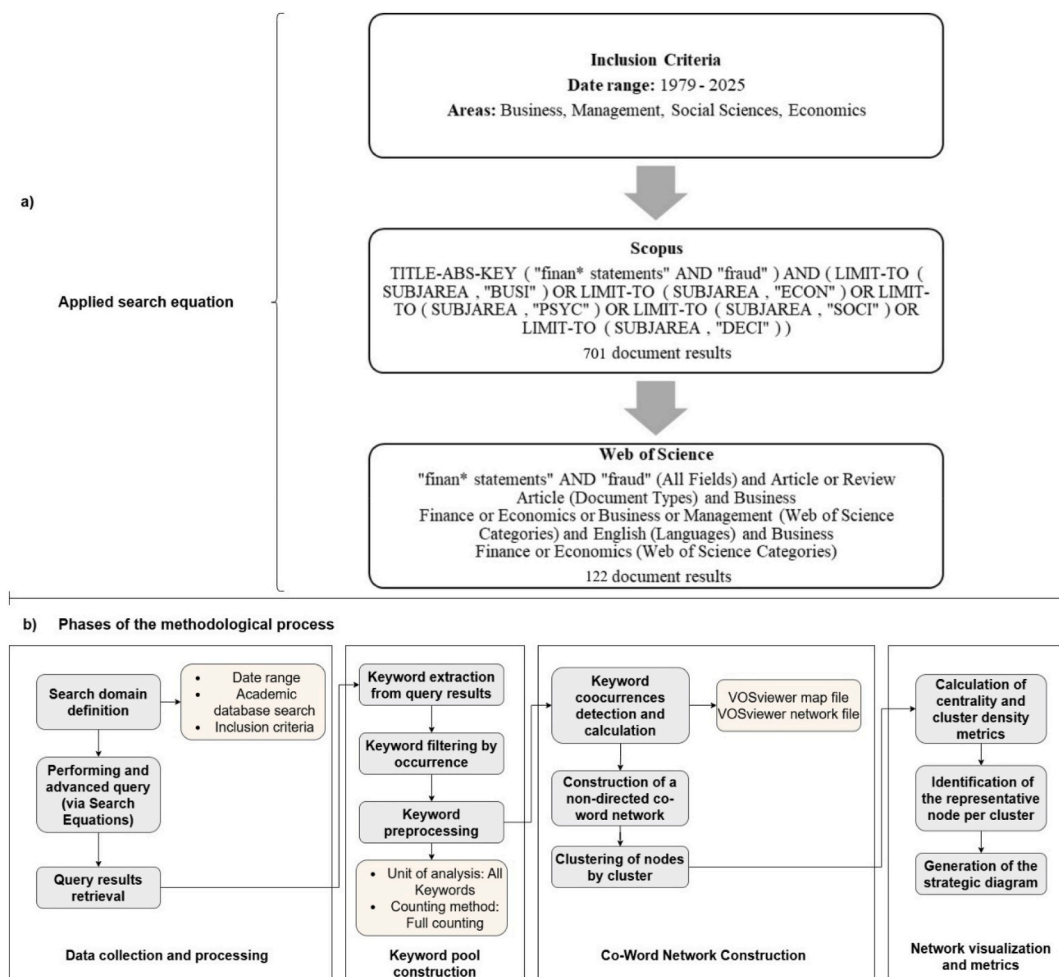


Fig. 1. Phases of the methodological process.

Source: Authors

2.2. Overview of bibliometric research methods

The bibliometric analysis encompasses an interdisciplinary study, which through mathematical and statistical techniques objectively evaluates the obtained literature, thereby avoiding the inherent subjectivity (Donthu et al., 2021). The article employs techniques of analyzing the number of publications/citations and scientific mapping for the study of co-occurrences, using the VOSviewer software in which the connections between elements are visualized (Van Eck & Waltman, 2010). Additionally, through the open-source Bibliometrix software based on R, publication statistics and international collaboration are analyzed (Aria & Cuccurullo, 2017). The data were cross-checked from the Scopus and Web of Science databases, stored in a Microsoft Excel spreadsheet. Subsequently, they were combined into a CSV file for further processing with VOSviewer and Bibliometrix software, in accordance with the methodology established by Echchakoui (2020).

Following the parameters of Batz et al. (2025) and Callon et al. (1991), a thematic network was generated from a normalized co-occurrence matrix of keywords using VOSviewer software. In the network, nodes represent author keywords, and links indicate their co-occurrence; larger nodes and thicker links reflect higher frequency and relationship strength, respectively. VOSviewer employs the Visualization of Similarities (VOS) approach to map keywords and cluster them. Nine thematic clusters were identified using the Newman and Girvan modularity algorithm, adjusting attraction and repulsion parameters to 2 and 1, and employing a resolution of 1, which provided better semantic segmentation.

From the generated clusters, two metrics were calculated: centrality and density; since VOSviewer does not provide these calculations, a script was used to extract the data and obtain these measures.

Centrality indicates the influence of a cluster within the entire field, based on its connections with other groups.

$$C_L = \sum_{i \in L} \sum_{j \notin L} w_{ij} e_{ij}$$

$$e_{ij} = \begin{cases} 1, & \text{if } w_{ij} > 0 \\ 0, & \text{otherwise} \end{cases}$$

Where, C_L is the centrality of cluster L, representing the measure of external connectivity of the cluster, that is, how much it relates to other clusters; $i \in L$, node i belonging to cluster L; $j \notin L$, node j that does not belong to cluster L, that is, it is outside the cluster; w_{ij} , weight of the link between nodes i and j , represents the frequency with which these keywords appear together; e_{ij} , binary indicator, takes the value 1 if there is a relationship between i and j (i.e., if $w_{ij} > 0$), otherwise it takes the value 0.

Density, on the other hand, measures the degree of internal connection within the cluster, reflecting its thematic maturity.

$$D_L = \frac{2E}{N(N-1)}$$

Where, D_L is the density of cluster L, measuring the degree of internal cohesion of the cluster; E , number of unique internal links within cluster L, i.e., how many connections there are between its own nodes; N , number of nodes (keywords) that make up cluster L; $N(N-1)/2$, maximum possible number of links between the N nodes (in an undirected graph).

Both metrics were represented in a Strategic Diagram. In it, clusters are distributed according to their centrality (X-axis) and density (Y-axis), divided into four quadrants: driving themes (upper right), specialized (upper left), emerging (lower right), and marginal (lower left). This visualization allows identifying current and future trends in the analyzed field.

3. Results and discussions

3.1. Analysis of general trends

Fig. 2 shows that in recent years there has been growing academic interest in the study of fraud in FSF.

Between the years 1979 and 1999 (20 years), academic production related to FSF was limited, with 41 documents recorded, with annual frequencies below five documents, preventing the identification of solid patterns from a quantitative perspective. Consequently, the present analysis focuses on the period between the years 2000 and 2025, with a total of 764 articles, where a consistent and systematic evolution of the literature is evident. This temporal cut-off more rigorously represents the dynamics of growth and their possible correlations with relevant events in the economic, regulatory, and corporate spheres. The figure associated with the analysis shows only this period, as it concentrates the highest volume of publications.

Starting in the year 2000, with a total of 6 documents, a period of progressive growth in research on fraud in FSF began. Between 2001 and 2004, the numbers remained in a range of 7 to 12 annual publications compared to 2009 with 20 documents, in the context following the global financial crisis. In 2011, 27 articles were recorded, reflecting an intensification of academic attention to the topic, which responds to increased awareness of the relevance of financial fraud and the influence of emblematic cases of corporate corruption, and the development and incorporation of analysis technologies by organizations (e.g., Enron financial fraud; *dot-com bubble*) regulatory bodies of the Securities and Exchange Commission (SEC). The period 2015–2019 shows a consolidation of academic interest, with figures ranging between 27 and 47 annual documents, years coinciding with the development of the FIFA corruption scandals and the Brazilian construction company Odebrecht. Finally, the five-year period 2020–2025 represents the peak of production, with 49 documents in 2020, 46 in 2021, 67 in 2022, 79 in 2023, 76 in 2024, and 84 in 2025, demonstrating a consolidation of

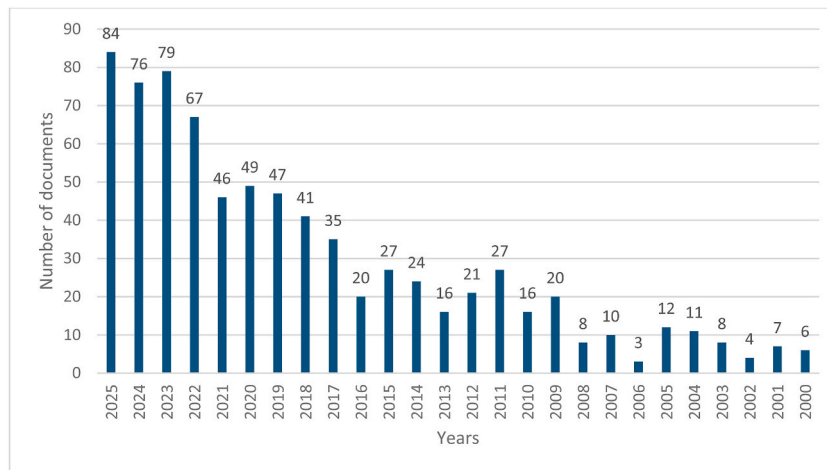


Fig. 2. General upward trend in publications.

Source: Authors

the area in the accounting and financial field.

The temporal dynamics identified in Fig. 2 call for a deeper examination of the recurrent interaction between major fraud scandals and regulatory response, as well as their role as a key driver of academic research on FSF. A seminal case of this dynamic is the Enron fraud and the subsequent enactment of the Sarbanes-Oxley Act (SOX) in the United States in 2002. These events transformed regulatory frameworks and business practices, and substantially reoriented the research agenda toward the analysis of internal control, audit committee oversight, and corporate governance as central mechanisms for fraud mitigation (Coates, 2007; Dechow et al., 2011).

Similarly, a more recent episode such as the collapse of FTX highlights a new phase in this relationship between fraud, regulation, and investigation. The significant increase in publications observed since 2022 (see Fig. 2) coincides with an intensification of regulatory debate and supervisory actions on crypto-assets at the international level (e.g., the approval of the Markets in Crypto-Assets Regulation by the European Parliament in April 2023), as well as with a period of accelerated development and adoption of Industry 4.0 technologies aimed at fraud detection and prevention. This pattern suggests that, in the current context, advanced digital technologies do not replace traditional control and governance mechanisms but rather establish themselves as a complementary and central axis in future research on FSF, as discussed in the following sections.

Table 1

The ten most cited studies.

Referencia	Título	Fuente	Total Citations (TC)	TC per Year	Normalized TC
1. Beasley (1996)	An Empirical Analysis of the Relation between the Board of Director Composition and Financial Statement Fraud	Accounting Horizons	2139	73,76	2,96
2. Farber (2005)	Restoring Trust after Fraud: Does Corporate Governance Matter?	Accounting Horizons	647	32,35	6,62
3. Beasley et al. (2000)	Fraudulent Financial Reporting: Consideration of Industry Traits and Corporate Governance Mechanisms	The Accounting Review	565	22,60	3,91
4. Efendi et al. (2007)	Why do corporate managers misstate financial statements? The role of option compensation and other factors	Journal of Financial Economics	481	26,72	5,95
5. Watts (2003)	Conservatism in Accounting Part II: Evidence and Research Opportunities	Accounting Horizons	425	19,32	4,23
6. Ravisankar et al. (2011)	Detection of financial statement fraud and feature selection using data mining techniques	Decision Support Systems	329	23,50	4,37
7. Kedia and Rajgopal (2011)	Do the SEC's enforcement preferences affect corporate misconduct?	Journal of Accounting and Economics	306	21,86	4,07
8. Feng et al. (2011)	Why do CFOs become involved in material accounting manipulations?	Journal of Accounting and Economics	297	21,21	3,95
9. Kirkos et al. (2007)	Data Mining techniques for the detection of fraudulent financial statements	Expert with Applications	295	16,39	3,65
10. Rezaee (2005)	Causes, consequences, and deterrence of financial statement fraud	Critical Perspectives on Accounting	256	12,80	2,62

Source: Authors

3.2. Analysis of influential literature

Table 1 presents the ten most influential articles in the field according to the number of citations. The studies address aspects of FSF from its origins, implications, preventive measures to aspects of corporate governance and its influence on the occurrence, detection, and prevention of FSF (Beasley, 1996; Beasley et al., 2000; Efendi et al., 2007; Farber, 2005; Feng et al., 2011; Rezaee, 2005). Additionally, three articles that focus on regulatory aspects of SF fraud are highlighted (Kedia & Rajgopal, 2011; Rezaee, 2005; Watts, 2003). Finally, the use of data mining techniques in fraud detection emerges (Kirkos et al., 2007; Ravisankar et al., 2011).

In the general trend described previously, most studies (9 out of 10) were conducted around the first decade of the 2000s; a period during which cases of fraud in SF committed by emblematic national companies with significant international recognition, such as¹: FlowTex (2000) in Germany; HIH Insurance (2001) in Australia; Parmalat (2003) in Italy; Equitable Life (2000) and MG Rover Group (2005) in the United Kingdom; CINAR (2001) and Nortel (2009) in Canada; Satyam (2009) in India; and Enron (2001), WorldCom (2002), Tyco (2002), Adelphia (2002), Healthsouth (2003), Freddie Mac (2003), American International Group (2005), Lehman Brothers (2008), Bernie Madoff (2008) in the USA, became publicly known with extensive media coverage.

3.3. Analysis of influential authors

Table 2 summarizes the main published authors. The number of studies published by the authors ranges between seven and five.

A review of the most prolific authors in the field of FSF reveals Joseph F. Brazel, Keith L. Jones, and Conan C. Albrecht; researchers who have published studies on the phenomenon of FSF from various perspectives. Brazel and Jones analyze that discrepancies between financial and non-financial measures are useful signals for identifying fraud risks in auditing. Additionally, Jones has delved into individual and attitudinal aspects related to financial manipulation. In turn, Jones evaluates the ability of accounting models to detect actual fraud and corrections of financial results.

On the other hand, Conan C. Albrecht focuses part of his work on technological trends and tools for fraud detection, highlighting the need for adaptive responses to new threats; statistical learning methods and data mining that have demonstrated a balance between effectiveness and interpretability are evaluated (Whiting et al., 2012). The above reflects that the research contains an interdisciplinary approach encompassing risk assessment and accounting modeling, incorporating technological tools into control processes, in order to generate strategies for the prevention and detection of FSF.

3.4. Research area analysis

The most prominent research areas in the topic of FSF revolve around organization and/or business, management, and accounting with 473 articles published, which introduces new emerging actors from a business and accounting perspective. Similarly, there is notable interest in economics, econometrics, and finance (363 articles), indicating that the economic analysis of fraud and its impact on society are topics of great interest. In parallel, social sciences (228 articles) consider fraud from sociological and psychological perspectives, while decision sciences (92 articles) and computer science (79 articles) fall within the development of models and technological tools for the detection and prevention of FSF.

3.5. Analysis of influential journals

Table 3 summarizes the top ten most influential journals, highlighting that FSF is an emerging research field, with each journal having a modest number of publications.

The journals adopt a pluralistic approach, encompassing various perspectives. The Journal of Financial Crime (36 articles) addresses financial crimes that include money laundering and corruption in the financial sector and presents prevention and detection measures to mitigate this type of fraud. The Journal of Business Ethics (18 articles) covers fraud-related topics from an ethical standpoint, examining its impact on the author's reputation and the conduct of the affected company. The Managerial Auditing Journal (14 articles) focuses on management auditing, with special emphasis on the detection and prevention of FSF. Journals such as Accounting Horizons and Issues in Accounting Education (12 articles each) adopt an educational and professional approach with practices and teaching methods for fraud prevention. The Accounting Review (11 articles) publishes articles from theoretical and methodological perspectives in accounting. Other journals like Auditing (10 articles) and Current Issues in Auditing (9 articles), by their very nature, publish articles centered on the detection and prevention of FSF. Finally, Contemporary Accounting Research and Critical Perspectives on Accounting (8 articles each) analyze the causes and consequences of fraud. These journals promote scientific knowledge by presenting research that contains best practices in fraud detection, contributing to the scientific community and especially to organizations requiring emerging solutions for the prevention and detection of fraud in financial reports.

3.6. Institutional analysis by influential countries

The top ten research institutions in terms of publications are listed in Table 4.

¹ The year of the scandal is written in parentheses.

Table 2

The ten most influential authors.

Authors	Institutions	Articles	Articles Fractionalized
Brazel J	North Carolina State University	7	1,95
Jones K	George Mason University	7	2,70
Albrecht C	Utah State University	6	1,62
Jamal K	University of Alberta	6	1,83
Na N	Xi'an Eurasia University	6	6,00
Riani M	Università di Parma	6	1,78
Smith J	Indiana University	6	2,83
Wang Y	Cheung Kong Graduate School of Business (CKGSB)	6	1,95
Johnson P	University of Minnesota	5	1,50
Kotsiantis S	University of Patras	5	1,50

Source: Authors

Table 3

The ten journals with the most articles.

Sources	Articles	Impact factor
Journal of Financial Crime	36	0.38
Journal of Business Ethics	18	2.62
Managerial Auditing Journal	14	0.76
Accounting Horizons	12	1.18
Issues in Accounting Education	12	0.42
Accounting Review	11	4.64
Auditing	10	1.88
Current Issues in Auditing	9	0.37
Contemporary Accounting Research	8	3.09
Critical Perspectives On Accounting	8	1.94

Source: Bibliometrix

Table 4

The ten institutions with the most articles.

Country	Institution	Total Articles (Institution)	Total articles (country)	Single Country Publication (SCP)	Multiple Countries Publication (MCP)
Unites States	University of Minnesota	9	105	100	5
	Arizona State University	8			
	Brigham Young University	8			
Singapore	Nanyang Technological University	3	54	53	1
Indonesia	Bina Nusantara University	8	44	43	1
India	Lovely Professional University	4	20	20	0
Malaysia	University Teknologi Mara	18	19	16	3
	Univ Teknol Mara	11			
United Kingdom	ondon School of Economics (LSE)	6	18	15	3
Romania	Bucharest University of Economic Studies	7	17	17	0
Canada	University of Alberta	5	16	16	0
Slovakia	Univ Econ Bratislava	10	16	15	1
Iran	Ferdowsi University of Mashhad	8	13	13	0

Source: Authors

Mara Technological University has the highest number of publications with research focused on information technology, finance, and accounting, among others, with programs addressing topics related to the detection and prevention of FSF in accounting, corporate finance, public finance, financial economics, risk management, and business strategies.

The scientific production by countries in the field of FSF reveals a heterogeneous geographical distribution of research activity. The United States leads the field with a production of 105 articles, reflecting its central role in both academic development and professional practice associated with accounting, auditing, and financial regulation (Free et al., 2020; Khan, 2017).

The bibliometric evidence shows that China and Indonesia rank second and third globally in scientific production, with 54 and 44 publications respectively. From an institutional perspective, emerging economies present accounting reporting and oversight frameworks with distinct characteristics, reflected in a heterogeneous adoption of international standards, variable audit quality, and comparatively more limited regulatory application (Ball, 2006; La Porta et al., 1998, 2006). These institutional traits shape environments where investor protection is deficient, and external control and compliance mechanisms tend to be less effective (La Porta

et al., 1998, 2006; Aguilera & Haxhi, 2019). As a result, managerial incentives for opportunistic earnings management are intensified, and the likelihood of manipulating financial statements (FS) increases (Dechow et al., 1996; Leuz et al., 2003). In this context, the growing academic attention in countries like China and Indonesia reflects their greater integration into global capital markets and, in turn, a higher underlying exposure to fraud risk and a rising demand for research aimed at addressing context-specific control and governance issues within the institutional framework.

On the other hand, bibliometric evidence also reveals relevant gaps in scientific production on FSF from certain regions. In the case of Latin America, despite problems such as tax evasion and public-private corruption having been widely documented in the region (Castañeda, 2016; Mejía-Flores et al., 2020), specific research on FSF remains limited. This mismatch reflects structural constraints in research capabilities and in the development of academic agendas specialized in forensic accounting.

Finally, the limited participation of Western European countries in this line of research can be interpreted in light of institutional environments characterized by higher levels of transparency, more robust investor protection, and more effective regulatory oversight. This contrast reinforces the interpretation that the intensity and orientation of research on fraud in FSF largely respond to institutional differences in accounting, regulatory, and governance systems, consistent with the higher output observed in emerging economies such as China and Indonesia.

3.7. Scientific mapping and keyword analysis

The study of keywords and emerging trends guides the evolution of knowledge and the development of research areas in the field of fraud in SF (Fig. 3). The analysis of keyword co-occurrence is a technique in the field of natural language processing (NLP) that reveals patterns of association between terms within a text corpus and infers semantic and contextual relationships between them (Donthu et al., 2021).

The thematic segmentation derived from the keywords breaks down the conceptual and structural relationships between the nodes associated with the study phenomenon. Additionally, each cluster reflects interrelated dimensions whose systematic exploration provides critical inputs to understand the dynamics of interaction and the need to address these approaches comprehensively. Fig. 3 identifies nine main clusters, which are analyzed in detail below.

Cluster 1: Governance and Analytical Factors in FSF

Fig. 4a shows the cluster grouping elements that provide analysis and decision-making within a company, considering numerical data, organizational structure, and management policies that influence financial outcomes.

The term "Fraud" (weight<Occurrences> 195) is the conceptual core in research on misconduct in SF, articulating central themes of

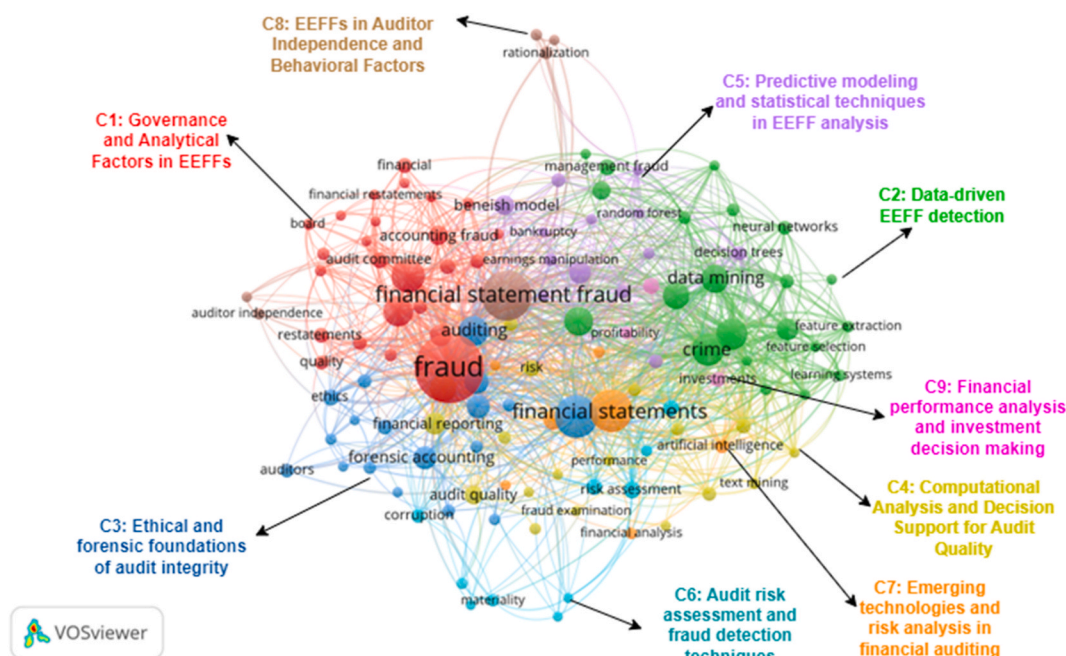
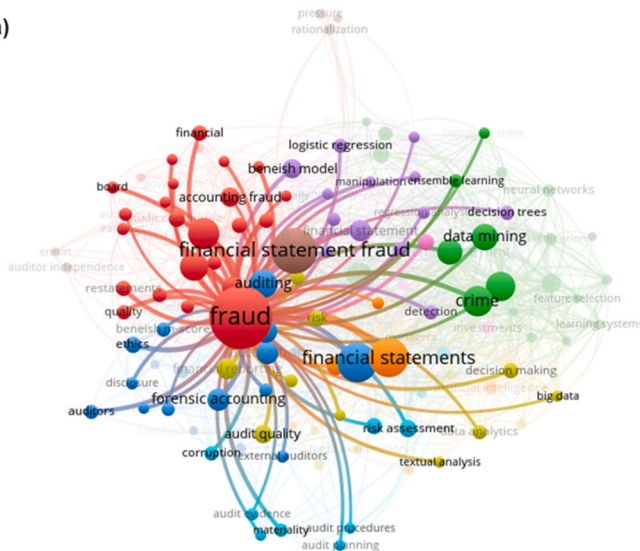


Fig. 3. Keyword co-occurrence map

Note. The color of each node indicates the cluster to which it belongs, and the size of the nodes is proportional to the degree centrality of the keywords, that is, the number of correlations each has, according to the VOS clustering algorithm. Source: Own elaboration.

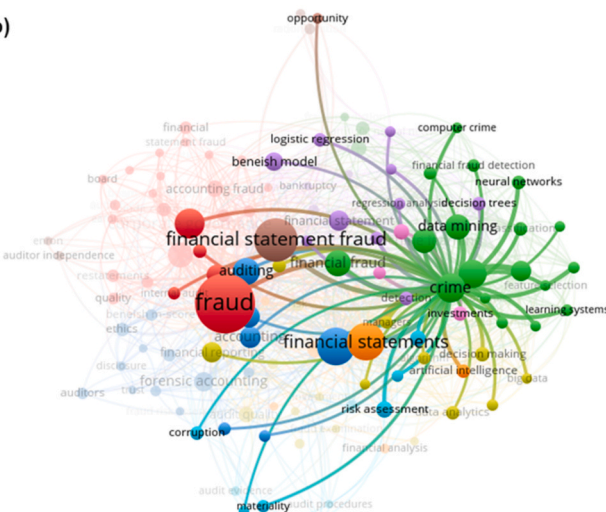
Source: VOSviewer

a)



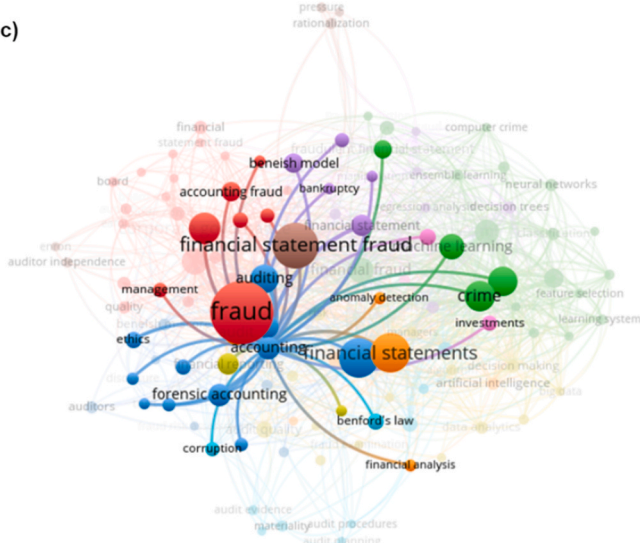
Keyword	Weight <Occurrences>
Accounting Fraud	17
Accounting Manipulation	5
Audit Committee	11
Audit Fees	7
Board	6
Board Of Directors	5
Corporate Fraud	10
Corporate Governance	46
Earnings	6
Earnings Management	37
Equity Incentives	5
Financial	10
Financial Restatements	5
Financial-Statements	9
Fraud	195
Fraud Diamond	5
Fraudulent Financial Reporting	20
Fraudulent Financial Statement	5
Governance	5
Internal Audit	7
Internal Control	9
Management	9
Quality	8
Red Flags	5
Restatements	10
Statement Fraud	5

b)



Keyword	Weight <Occurrences>
Classification	9
Computer Crime	5
Crime	45
Data Mining	31
Deep Learning	6
Economics	5
Ensemble Learning	7
Feature Extraction	6
Feature Selection	5
Finance	40
Financial Fraud	35
Financial Fraud Detection	7
Financial Fraud Detections	5
Financial Statement Fraud Detection	5
Financial Statement Frauds	20
Fraud Prediction	5
Fraudulent Financial Statements	15
Learning Systems	6
Machine Learning	30
Management Fraud	12
Neural Networks	8

c)



Keyword	Weight <Occurrences>
Accounting	22
Auditing	38
Auditor	7
Auditors	9
Beneish M-Score	8
Benford's Law	7
Business	7
Disclosure	6
Ethics	11
External Auditors	6
Financial Crime	5
Financial Statements Fraud	7
Forensic Accounting	22
Fraud Detection	74
Fraud Prevention	10
Fraud Risk Assessment	5
Fraud Triangle	24
Trust	5

(caption on next page)

Fig. 4. Expanded view of clusters 1, 2, and 3

Note: a) Cluster 1: Governance and analytical factors in SF; b) Cluster 2: Detection of FSF based on data; c) Cluster 3: Ethical and forensic foundations of audit integrity. The color of each node represents the cluster to which it belongs, and the size of the nodes is proportional to the degree centrality of the keywords, that is, to the number of correlations each has, according to the VOS clustering algorithm. Source: Own elaboration.

"Corporate Governance" (46) and "Earnings Management" (37); connections that show weaknesses in governance and deliberate earnings adjustments are determining factors in the emergence of fraud. This approach is developed by Hilal et al. (2022) where they present an analysis of anomaly detection techniques applied to financial fraud, highlighting its prevalence and complexity. For their part, Suljić et al. (2025) demonstrate the usefulness of data mining to identify irregularities in audited SF, emphasizing the potential of state audits. Additionally, Nyakarimi (2022) analyzes that in the African banking sector there is a direct relationship between earnings manipulation practices and structural failures in governance. Finally, Shah and Wan (2024) mention that financial integration provides scenarios of opportunistic earnings management in emerging markets.

The control mechanisms related to "governance" (5), the "audit committee" (11), and "internal audit" (7) impact the supervision and detection of fraud. Similarly, Kalina et al. (2022) argue that a corporate risk management system strengthens internal controls against financial threats. In this regard, Mabelane et al. (2022) maintain that the use of auditing and, at the same time, machine learning (ML) is necessary to detect anomalies in local government environments. Lokanan et al. (2019) apply ML techniques to assess the credit quality of SF in Vietnamese companies. Meanwhile, Rashid et al. (2023) delve into the analysis of auditors' responses to fraud, revealing the importance of strong institutional structures.

Regarding incentive schemes, "equity incentives" (5) and employee benefits are considered critical factors in the occurrence of fraud. Chen et al. (2023) analyze that some financial incentives directly influence corporate disclosure, increasing the risk of manipulation. Similarly, Vismara (2022) examines new financing mechanisms (crowdfunding), and in turn, the alteration of incentives and risks associated with financial transparency.

"Restatements" (10) are signs of financial fraud. Dutta et al. (2017) They explore data mining techniques and identify patterns associated with the need for financial corrections. On the other hand, Hajek and Henriques (2017) combine text analysis and accounting variables to detect patterns in annual reports that predict the occurrence of fraud with high accuracy, while Le and Moore (2023) analyze the relationship between audit quality and the occurrence of restatements and earnings management. For their part, Wang and Wu (2011) examine that restatements reflect structural weaknesses in the quality of financial reporting in China. In turn, specialized audit functions are key. Dhamija and Nayyar (2024) argue that failures in the governance of startups allow financial manipulation, and Liu et al. (2023) investigate that changes in accounting disclosures anticipate fraud; and Mandal and S (2025) identify good preventive practices from the perspective of specialized auditors.

Finally, the temporal approach reveals that traditional topics of "internal audit" (7), "governance" (5), and "equity incentives" (5) were more developed before 2017, while new research lines emerge focused on "red flags" (5), "fraudulent financial statement" (5), and "quality" (8); trends that are captured by Shahana et al. (2023), who present a systematic review identifying emerging indicators to address accounting fraud with a more updated perspective.

Cluster 2: FSF Detection Based on Data

Fig. 4b shows cluster 2, which groups the analytical methodologies and tools used to identify patterns and trends within FS, enabling a more accurate assessment of economic health and company performance based on available information. Transparency in FS is crucial to meet stakeholder expectations, but these reports are often subject to intentional manipulation. At a conceptual level, the term "crime" (weight <Occurrences> 45) encompasses deliberate errors and bad practices in organizations; an approach that is theoretically connected to criminology, where De La Torre Lascano and Quiroz (2023) it is analyzed that financial crime takes new forms in digital environments and is associated with disruptive events of the pandemic.

Similarly, a branch is "computer crime" (5), which emphasizes the digital dimension of fraud. Ali, Nipu, and Khan (2023) develop a model based on XGBoost to detect financial fraud, demonstrating the effectiveness of ensemble learning against computer crimes in corporate environments, and Achenbach (2021) evidence that unprotected digital systems grant unauthorized access that facilitates accounting manipulation, and exemplifies the concept of "digital opportunity" as a facilitating condition for fraud, shaping the analysis of technologically mediated financial crimes.

Within the corporate environment, the term "financial fraud" (35) becomes relevant when dealing with manipulation by executives motivated by compensation. It is representative, the study of Achakzai and Peng (2023) considering that adverse economic conditions or bonus schemes can lead to deliberate distortions of the SF. The most evident manifestation of these practices is presented in the "fraudulent financial statements" (15), reports that include intentional material errors. In this aspect, Dhamija and Nayyar (2024) illustrate that poor corporate governance culminates in fraudulent financial statements; a paradigmatic case in the context of startups.

On the other hand, the pressures of "management fraud" (12) and the conditions of "economics" (5) are the structural drivers of fraud. Nguyen et al. (2020) They explain that market expectations and economic incentives influence the level and quality of disclosed information, linking the economic environment with the propensity for fraud. The use of "data mining" (31) and "machine learning" (30) techniques gains prominence through the identification of anomalous patterns in large volumes of accounting data (Gupta & Mehta, 2021).

Revisankar et al. (2011) highlight that techniques such as "feature selection" (5) and "classification" (9) show indicators of atypical levels of income or cash flows to distinguish fraudulent companies from legitimate ones. To improve detection accuracy, more complex

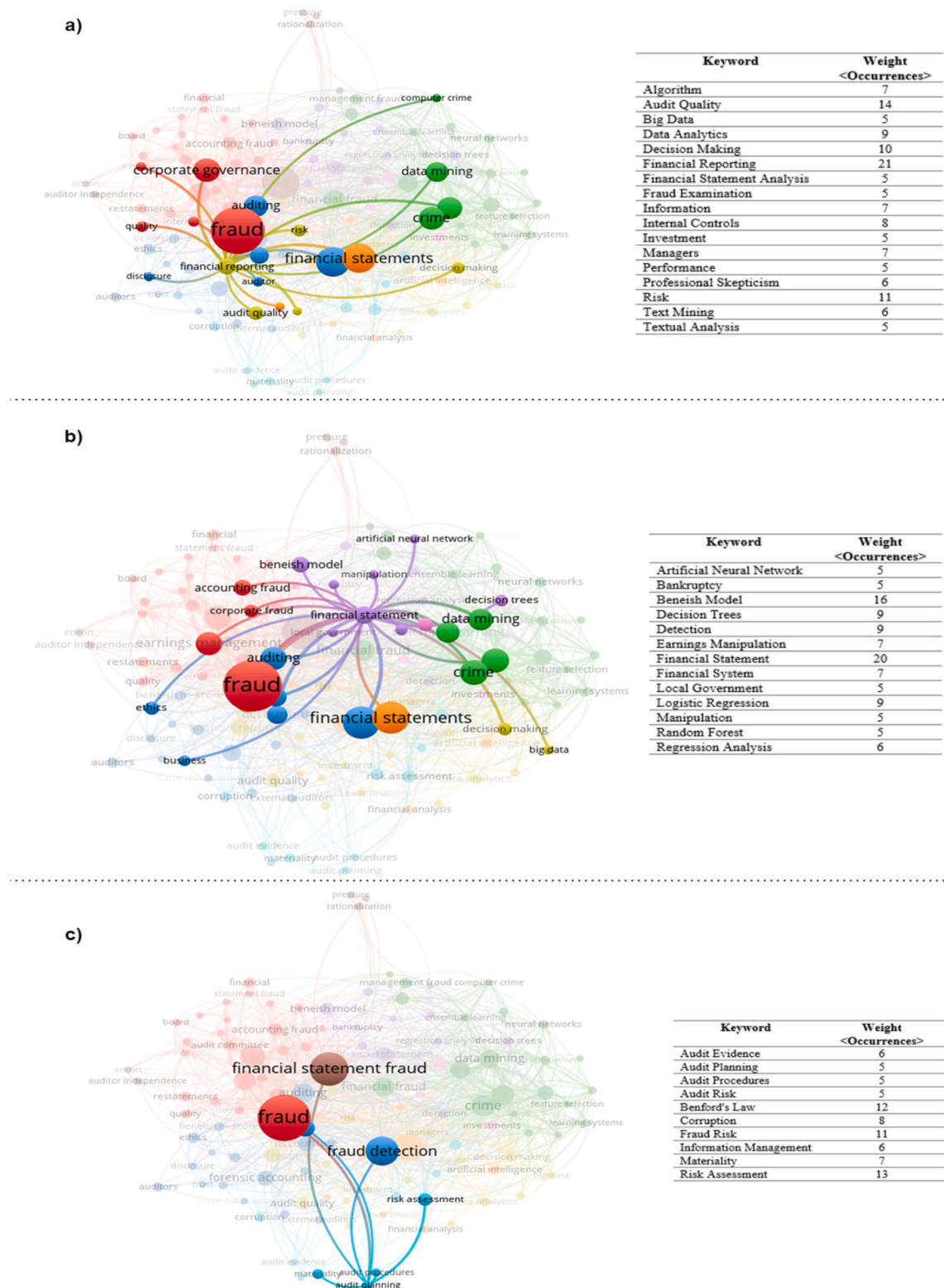


Fig. 5. Expanded view of clusters 4, 5, and 6

Note: a) Cluster 4: Computational analysis and decision support for audit quality; b) Cluster 5: Predictive modeling and statistical techniques in the analysis of SF; c) Cluster 6: Audit risk assessment and fraud detection techniques. Source: Authors.

models are integrated through "ensemble learning" (7) and neural networks; integration developed by [Chen et al. \(2014\)](#) combining models to enhance predictive capability against accounting fraud. Finally, "financial fraud detection" (7) focuses on practical tools for auditors and regulators. It is there, where [Ramírez-Alpízar et al. \(2020\)](#) mention that the area under the curve (AUC) helps calibrate risk thresholds and improve the supervision of FSF.

Cluster 3: Ethical and Forensic Foundations of Audit Integrity

[Fig. 4c](#) shows cluster 3, which groups the principles and practices of the audit process to protect the integrity of financial information. The terms "Accounting" (weight<Occurrences> 22) and "Auditing" (38) affirm the institutional and ethical principles of SF. In this context, [Estep et al. \(2024\)](#) they explain that technological tools impact the accuracy and reliability of financial reports and audit engagements, such as automated audit software like ACL Analytics and IDEA, big data and data mining platforms like RapidMiner, and AI and ML solutions (GLAi, IBM Watson, SAS Fraud Management, and DataRobot), emerging technologies like blockchain (Chainalysis, CipherTrace), and forensic analysis tools (Tableau, Benford's Law). These tools enable more precise and proactive detection of FSF ([Nigro, 2025](#)).

Additionally, the audit presents the variables that generate the "fraud triangle" (24), such as: pressure, opportunity, and rationalization. The concept of "trust" (5) aims to maintain confidence in organizations and their reports. Likewise, [Abdolmohammadi and Owhoso \(2000\)](#) they emphasize that the ethics of auditors influence the detection of fraud and the integrity of reports. In the field of financial crime, "financial crime" (5) and "financial statements fraud" (7) reflect the negative effects of fraud on public trust ([Ramírez-Alpízar et al., 2020](#)).

Regarding the topic of "forensic accounting" (22), as a technique it reconstructs altered records, and "fraud detection" (74) encompasses statistical methods that identify fraud. [Kranacher and Riley \(2019\)](#) offer a framework in the application of forensic auditing. The "Beneish M-Score" (8) and "Benford's Law" (7) models identify numerical anomalies in FSF. [Mohamad Kamal et al. \(2016\)](#) mention that the Beneish model is effective in detecting fraud in companies listed on the stock exchange. The "fraud risk assessment" (5) evaluates exposures to errors and fraud. What makes it relevant for [Ocampo \(2023\)](#) is the use of AI, improving fraud detection and the reliability of SF, as well as emerging technologies like deep learning and forensic analysis in blockchain that detect complex patterns and track transactions with high precision ([Nigro, 2025](#); [Xiuguo & Shengyong, 2022](#)). Finally, "fraud prevention" (10) strategies include strengthening internal controls and training employees in ethics to minimize fraud risks. [Chen et al. \(2024\)](#) present an approach to identifying erroneous accounts in SF based on ontologies.

Cluster 4: Computational analysis and decision support for audit quality

[Fig. 5a](#) shows cluster 4, which groups analytical techniques based on advanced technologies during the audit process. On one hand, [Hajek and Henriques \(2017\)](#) they apply computational analysis by combining text mining and accounting variables to detect FSF patterns:

"Financial reporting" (weight<Occurrences> 21) and "financial statement analysis" (5) generate and review accounting information, while "audit quality" (14) and "internal controls" (8) ensure the reliability of the report. [Brasel et al. \(2016\)](#) address tools that influence the perception of auditor responsibility. According to [Al-Okaily and Al-Okaily \(2024\)](#), the rise of "big data" (5) and "data analytics" (9) optimizes anomaly detection and improves the quality of financial decisions; advancements that are complemented by the use of "algorithms" (7), "text mining" (6), and "information" (7) extracted from unstructured data; in this way, these methods find patterns for manual and automated reviews ([Blanco et al., 2023](#)).

Even traditional processes of evaluating "investment" (5) and "performance" (5) are being redefined through computational approaches and techniques. [Ashtiani and Raahemi \(2022\)](#) They highlight that these techniques better estimate risks and the financial health of companies; however, they require human judgment to interpret the results. "Decision making" (10) and "professional skepticism" (6) play a key role, as managers adjust risk thresholds based on algorithmic outcomes and their own assessments ([Sequeira et al., 2024](#)).

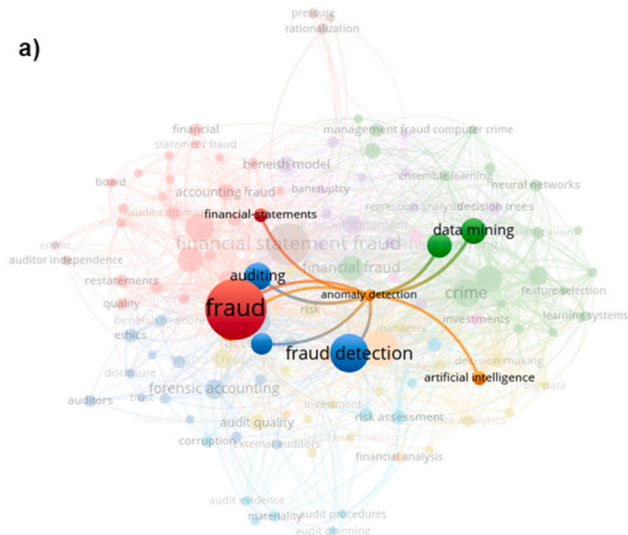
The frameworks of "risk" (11) help identify areas with the highest vulnerability to fraud, where specific procedures should be applied. [Duan et al. \(2024\)](#) propose learning models that improve this identification. Similarly, "fraud examination" (5) represents the detection of irregularities in internal control and governance. [Lokanan \(2014\)](#) exemplify the detection of fraudulent behaviors based on organizational patterns. Finally, "textual analysis" (5) uncovers linguistic signals of fraud (evasive or ambiguous language). [S. Gupta and Mehta \(2021\)](#) demonstrate that textual analysis complements numerical analysis to identify fraudulent reports; the cluster integrates computational tools with professional judgment to strengthen the integrity of SF and improve the effectiveness of data-based audits.

Cluster 5: Predictive Modeling and Statistical Techniques in the Analysis of SFs

[Fig. 5b](#) shows cluster 5, which groups together the methodologies and analytical approaches that use statistical models and predictive algorithms to anticipate trends and financial behaviors, thereby enhancing analytical capabilities and decision-making in assessing the financial health of companies. On the other hand, [Ali, Nipu, and Khan \(2023\)](#) employ predictive modeling and advanced statistical techniques such as XGBoost, achieving 93.66% accuracy in fraud detection, surpassing classical models like SVM and logistic regression.

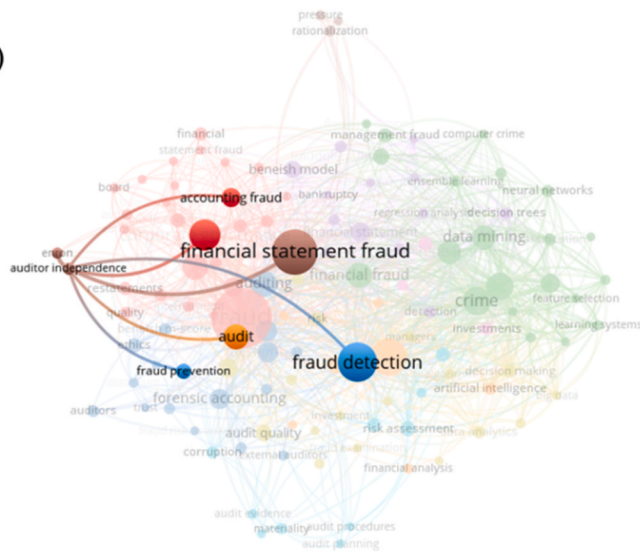
The cluster also addresses "earnings manipulation" (weight<Occurrences>7) and discrepancies in "financial statement" (20), which

a)



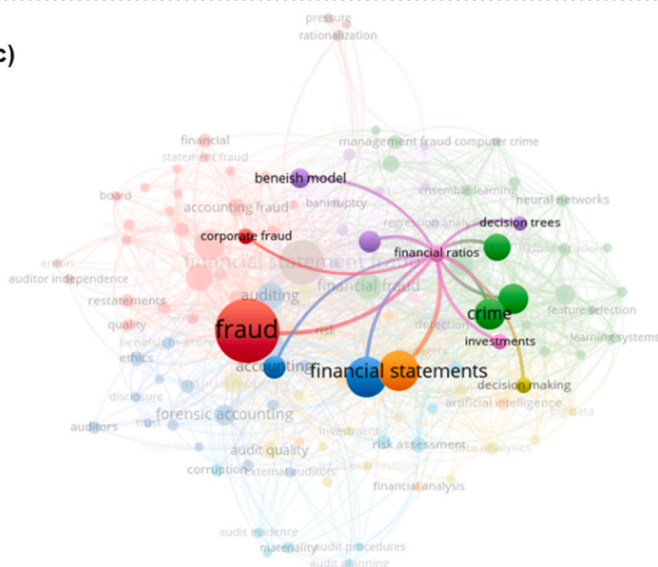
Keyword	Weight <Occurrences>
Anomaly Detection	6
Artificial Intelligence	9
Audit	29
Blockchain	5
Creative Accounting	16
Financial Analysis	6
Financial Crimes	5
Financial Statements	76

b)



Keyword	Weight <Occurrences>
Auditor Independence	6
Enron	5
Financial Statement Fraud	103
Opportunity	5
Pressure	6
Rationalization	7

c)



Keyword	Weight <Occurrences>
Financial Ratios	12
Investments	10
Profitability	7

Fig. 6. Expanded view of the subgrouping in groups 7, 8, and 9

Note: a) Cluster 7: Emerging technologies and risk analysis in financial auditing; b) Cluster 8: SF in the auditor's independence and behavioral factors; c) Cluster 9: Analysis of financial performance and investment decision-making. Source: Authors.

reflect deliberate alterations in reported results. [Narsa et al. \(2023\)](#) These distortions are analyzed at the core of financial scrutiny. Statistical methods such as "logistic regression" (9), "regression analysis" (6), and the "Beneish model" (16) transform these alterations into scores that anticipate fraud; the Beneish M-score model, for example, is calibrated to identify companies with a high probability of manipulating their FSF. [Mohamad Kamal et al. \(2016\)](#) This approach is represented in public companies in Malaysia.

ML complements these traditional methods with "decision trees" (9) and "random forest" (5), which hierarchically segment companies based on risk indicators. [Ali et al. \(2022\)](#) They highlight that these models increase accuracy in classifying fraudulent entities. Other algorithms, such as the "artificial neural network" (5), detect nonlinear patterns and subtle signals of inaccuracy. [Omar et al. \(2017\)](#) They demonstrate their usefulness in anticipating fraud based on complex relationships between variables. Beyond the corporate environment, these models predict critical events of "bankruptcy" (5) and allow the modeling of the financial system, reflecting their systemic relevance. [Akhigbe et al. \(2005\)](#) They show that accounting errors have cascading effects on financial markets.

These techniques are applied to the public sphere, the reference to "local government" (5) highlights that predictive models help identify hidden fiscal risks. The common conceptual basis is "manipulation" (5), which ties all these methodologies together in order to separate real performance from that which is intentionally reported in a misleading manner. [Anning and Adusei \(2022\)](#) delve deeper into this behavioral dimension of fraud. The cluster demonstrates a comprehensive approach ranging from classical statistics to ML, strengthening the ability to issue early warnings and forensic confirmations regarding the integrity of the SF.

Cluster 6: Audit Risk Assessment and Fraud Detection Techniques

[Fig. 5c](#) shows cluster 6, which covers alert signals, analytical tools, and professional judgment to detect accounting anomalies. This cluster highlights that the authenticity of the figures reported in the SF depends on the audit process (planning, execution, reporting, and monitoring). The cycle begins with "audit planning" (weight<Occurrences> 5) and extends through "audit procedures" (5), culminating in the collection of "audit evidence" (6); stages that establish the traceability of financial information from the moment of accounting record to the issuance of final reports. [Brasel et al. \(2016\)](#) state that this sequence values the robustness of the SF. During planning, auditors assess "audit risk" (5), adjusting the thresholds of "materiality" (7) and critical areas of analysis, where errors or fraud may arise, intersecting with the identification of "fraud risk" (11) and "corruption" (8), factors that represent threats that must be mitigated. [Bader et al. \(2024\)](#) offer a systematic approach to recognizing the risks present in the SF of the industrial sector.

The "risk assessment" tools (13) prioritize the most exposed accounts, while proper "information management" (6) ensures the integrity and traceability of data throughout the audit process. [Găbudeanu et al. \(2021\)](#) emphasize that information flows maintain accurate evidence and protect against manipulation. Regarding analytical techniques, the application of "Benford's Law" (12) stands out, identifying numerical irregularities through digit analysis, accounting entries that deviate from the expected statistical pattern, generating alert signals ([Mohamad Kamal et al., 2016](#)); and [Nigro \(2025\)](#) highlights its use alongside tools such as Tableau, regression analysis, text mining, and network analysis that strengthen the forensic detection of complex frauds. Audit procedures translate these signals into concrete evidence through confirmations, certifications, substantive analysis, and iterative evaluations. Thus, professional judgment is combined with quantitative tools to ensure that the SF faithfully reflect the economic reality of organizations. [Wan-Hussin et al. \(2021\)](#) reinforce that these practices strengthen audit standards and transparency.

Although it is highlighted that, while the cluster does not explicitly mention 4.0 technologies, concepts such as "fraud risk," "risk assessment," and "information management" reflect areas where these technologies are already being applied. Studies like those by [Ali, Nipu, and Khan \(2023\)](#) and [Hajek and Henriques \(2017\)](#) show that ML techniques and text mining improve fraud detection through predictive models. For its part, Benford's law has been automated in tools such as Tableau and combined with regression analysis and text mining to strengthen forensic auditing ([Nigro, 2025](#)); applications that demonstrate the potential of 4.0 technologies in the effectiveness and precision of risk assessment and procedures in detecting FSF. The cluster allows us to observe the importance of applying 4.0 technologies to generate efficient controls that protect the organization's finances and, in turn, reflect reliable information about the economic reality, which leads to decisions being made with real figures.

Cluster 7: Emerging Technologies and Risk Analysis in Financial Auditing

[Fig. 6a](#) shows cluster 7, which covers advanced concepts related to the integration of emerging technologies and analytical techniques in auditing and analysis of FRS with an emphasis on anomaly detection and fraud prevention.

The cluster focuses on technology that redefines the supervision and validation of SF. First, "financial statements" (76) constitutes the structural axis of the cluster and is the object that all these technologies aim to protect. On the other hand, "anomaly detection" (weight<Occurrences> 6) represents computational tools such as GLAI, IBM Watson, DataRobot, and AutoML, along with deep learning models (CNN, LSTM, RNN), and algorithms like XGBoost, Random Forest, clustering, association rules, and expert systems, which identify irregular patterns in large volumes of data, detecting errors or fraud early. [Ahmed et al. \(2016\)](#) explain that these techniques are alert sensors to detect atypical financial behaviors. [Kumar et al. \(2022\)](#) show that "artificial intelligence" (9) increases predictive accuracy in scenarios of complex financial manipulation; [Leal et al. \(2020\)](#) detail that the term "audit" (29) is a guarantor of financial reliability and has been transformed by digital tools, and in this sense, the role of auditing remains central.

On the other hand, [Amponsah et al. \(2022\)](#) they highlight that "blockchain" (5) provides an immutable and decentralized transaction record, enhancing audit traceability in digital environments. [Blanco et al. \(2023\)](#) they emphasize that "creative accounting" (16) complies with accounting regulations but manipulates results to present a desired image, where such strategies require refining auditing and control procedures. "Financial analysis" (6) are liquidity/profitability indicators that reveal warning signs of potential manipulations. [Jaswadi et al. \(2024\)](#) they explain that this type of analysis anticipates fraud even before regulators intervene. In parallel, "financial crimes" (5), such as money laundering or embezzlement, demand more robust detection techniques. [Nicholls et al. \(2021\)](#) in their review of financial cybercrime, they demonstrate the crucial role that deep learning techniques play in threat detection.

Cluster 8: The SF in auditor independence and behavioral factors

Fig. 6b shows cluster 8, which groups the elements that impact the objectivity and impartiality of the auditor in the FSF.

The cluster focuses on "auditor Independence" (weight<Occurrences> 6); which constitutes a person who is independently linked to the company, in order to minimize accounting manipulation [Alleyne and Howard \(2005\)](#). They highlight that this independence is compromised by organizational pressures or conflicts of interest, undermining the auditor's effectiveness. The "Enron" scandal (5) is a case that became a historical milestone triggering profound reforms under the Sarbanes-Oxley Act [Petra and Spieler \(2020\)](#). They explain that the close relationship between auditors and management at Enron exposed structural flaws that continue to raise concerns in regulation and professional auditing practices. On the other hand, "financial statement fraud" (103) represents the most severe manifestation of the collapse of professional independence and organizational ethical values ([An & Suh, 2020](#); [Ravisankar et al., 2011](#)).

Simultaneously, the fraud triangle provides a complementary psychological perspective to technical analysis; and is articulated through three components: "opportunity" (5), "pressure" (6), and "rationalization" (7). *Opportunity* refers to failures in internal control that enable fraud; *pressure* alludes to personal/work motivations in financial difficulties that drive dishonest decisions; and *rationalization* refers to the mental mechanisms that allow perpetrators to morally justify their illicit acts. [Hudayati et al. \(2022\)](#) and [Marquart and Thompson \(2024\)](#) emphasize that these variables are increasingly integrated into fraud detection and prevention models, revealing a behavioral shift in accounting research. In this sense, [Ali, Nipu, and Khan \(2023\)](#) and [Hajek and Henriques \(2017\)](#) highlight that financial, textual, and non-financial variables are increasingly integrated into advanced fraud detection and prevention models, improving accuracy and adaptability. This integrative approach, combining institutional, regulatory, and psychological factors, expands the understanding of financial fraud and guides the design of more effective controls tailored to organizational realities.

Cluster 9: Analysis of financial performance and investment decision-making

Fig. 6c shows cluster 9, which groups the approaches, tools, and metrics used to evaluate the profitability, solvency, and financial efficiency of organizations, providing strategic financial investment information by analysts, managers, and stakeholders.

The cluster presents the articulation between financial ratios, investments, and profitability as tools for the analysis of SF and strategic decision-making in investment contexts. The term "financial ratios" (weight<Occurrences> 12) are indicators that evaluate aspects of liquidity, solvency, operational efficiency, and profitability of an entity, which provides the financial analysis of the auditor against fraud. [Al-Okaily and Al-Okaily \(2024\)](#) show that financial ratios are integrated with data analysis technologies to improve the quality of financial decisions (ACL Analytics, IDEA, Tableau, RapidMiner, etc.) ([Nigro, 2025](#)), [Kootanaee et al. \(2021\)](#) y [Lokanan et al. \(2019\)](#) show that data analysis technologies and hybrid models improve the quality of financial decisions through early detection of anomalies and fraud. While [Stalebrink and Sacco \(2007\)](#) emphasize their value even in governmental contexts, by identifying signals of rationalization of accounting fraud.

On the other hand, the concept of "investments" (10) primarily addresses from the perspective of efficient resource allocation, aiming to maximize expected returns. [Roszkowska \(2021\)](#) and [Roychowdhury et al. \(2019\)](#) explore how the quality of financial reporting and transparency in disclosure directly influence investment decisions and market confidence. Finally, "profitability" (7) represents a business performance criterion, closely linked to the previous two terms; it evaluates the company's ability to generate profits in relation to revenues, assets, or capital. [Díaz et al. \(2021\)](#) and [Tang et al. \(2023\)](#) demonstrate that a profitability analysis detects early warning signs of potential financial fraud; the cluster helps guide informed investment decisions and project future profitability.

3.8. Content analysis

Content analysis allows potential researchers in this field to identify knowledge gaps in the literature ([Donthu et al., 2021](#)). The content analysis was conducted based on the most cited articles in the related literature, and for this purpose, the most cited articles were selected, undergoing a thorough analysis process to identify recurring themes emerging in the texts. Subsequently, these themes were classified and grouped into subtopics, facilitating a more detailed understanding of the predominant research areas. This analytical approach was based on a total of 47 unique scientific articles and was complemented by the creation of a co-occurrence map, which enhanced the visualization of the relationships between the main identified themes.

The analysis revealed three themes: 1) Advanced methods and techniques for fraud detection associated with clusters 4, 5, and 6; 2) Impact of crises and specific sectors on the manipulation of FS associated with clusters 1, 2, and 7; and 3) Emerging technologies and their impact on accounting transparency associated with clusters 3, 8, and 9; a methodological approach that facilitated a global and systematic view of the current state of research.

3.8.1. **Topic 1: advanced fraud detection methods and techniques**

In the context of FSF, the focus is on the development and application of advanced methods to detect and prevent accounting and financial manipulations. Researchers and academics explore various analytical and technological strategies designed to identify irregular patterns in financial information that indicate fraudulent practices (Alden et al., 2012; Bakumenko & Elragal, 2022; Jan, 2021; Kirkos et al., 2007; Perols, 2011; Ramírez-Alpízar et al., 2020; Shahana et al., 2023; Yao et al., 2018), ranging from inflating revenues, concealing liabilities, or altering figures to improve a company's financial appearance.

3.8.1.1. Subtopic 1.1: detection of outliers and elimination of arbitrary significance levels. The identification of outliers in financial data is of utmost importance in detecting potential FSF cases as anomalies indicate instances of manipulation (Ravisankar et al., 2011; Rizki et al., 2017). To avoid erroneous or biased conclusions, researchers use advanced statistical techniques of multivariate analysis, clustering algorithms, and ML models that develop objective criteria to differentiate between normal fluctuations and suspicious behaviors (Aboud & Robinson, 2022; An & Suh, 2020). The use of K-NN algorithms, principal component analysis (PCA), and anomaly detection models, including Isolation Forest and One-Class SVM (An & Suh, 2020; Jan, 2021; Yao et al., 2018), adjust detection thresholds to the characteristics of financial data and their economic environment (Hajek & Henriques, 2017).

3.8.1.2. Subtopic 1.2: joint learning approach to assess the risk of FSF. Examine the potential of ensemble learning to improve the assessment of FSF risk through the integration of multiple predictive models to obtain more accurate and robust results in detecting accounting manipulations (Papík & Papíková, 2021). Researchers have combined ML models, including decision tree classifiers, neural networks, and ensemble methods (Bagging, Boosting, Random Forests), with the aim of enhancing predictive capability and generalization of the models in question (Rizki et al., 2017). This approach captures the complexity and interactions among financial variables, including the correlation between income and expenses, anomalous patterns in cash flow, and inconsistencies in ratios (Tang et al., 2018; Wang et al., 2023). Methodologies for training, validating, and optimizing models have been developed using cross-validation techniques and hyperparameter tuning to maximize accuracy and minimize the risk of overfitting (Phong et al., 2024; Riskiyadi, 2024). Additionally, strategies to improve model interpretability have been investigated, evaluating the effectiveness of the approach in diverse business contexts and its implications for early detection and mitigation of financial fraud (Achakzai & Peng, 2023).

3.8.1.3. Subtopic 1.3: dataset for fraud detection and hierarchical tree of FSF. The existing literature highlights the importance of a systematic and data-driven approach for the effective identification of anomalies in SF (Achakzai & Peng, 2023; Ashtiani & Raahemi, 2022; Hilal et al., 2022; Hogan et al., 2008; Kootanaee et al., 2021; Lokanan et al., 2019; Ramírez-Alpízar et al., 2020; Rezaee, 2005). A framework is proposed that integrates the creation of an efficient dataset with the application of a hierarchical fraud tree, which optimizes the classification and detailed analysis of financial irregularities (Achakzai & Peng, 2023; Hilal et al., 2022). An effective dataset must be of high quality, representative, and accessible; incomplete or erroneous data lead to incorrect conclusions. The collection of financial data is fraught with difficulties (lack of transparency and heterogeneity of sources), to ensure the feasibility of this data, cleaning and normalization techniques must be applied (Bakumenko & Elragal, 2022). On the other hand, the construction of a hierarchical fraud tree organizes types of fraud based on their common characteristics, thereby optimizing detection algorithms and improving the response to emerging threats (Encalada, 2023; Rodríguez, 2023).

3.8.2. **Topic 2: the impact of crises and specific sectors on the manipulation of SFs**

This topic examines economic crises and specific situations related to financial statement (SF) manipulation practices in different industrial sectors. The impact of the COVID-19 pandemic crisis led to increased pressure on companies to alter their financial information to maintain the appearance of stability or profitability, especially in vulnerable sectors affected by economic crises (Jaswadi et al., 2024; Shahana et al., 2023); a situation that prompted the study of ways in which local governmental organizations can improve their internal control mechanisms to prevent and detect FSF (Blanco et al., 2023; De La Torre Lascano & Quiroz, 2023).

Research on the topic has addressed: (1) understanding the underlying causes of FSF (Lisic et al., 2015; Sun et al., 2023; Wu et al., 2022); (2) institutional aspects of the weakness in governance structures and the absence of audit committees (Cohen et al., 2022; Ghafran et al., 2022; Wan-Hussin et al., 2021); (3) financial characteristics associated with fraud through sales revenue and discretionary reserves (Gui et al., 2024; Lu et al., 2024; Zheng et al., 2024); (4) characteristics of CEOs involved in financial fraud (Biggerstaff et al., 2015; Choi et al., 2024; Heyden et al., 2023); and (5) detection of FSF (Ashtiani & Raahemi, 2022; Ramírez-Alpízar et al., 2020).

Researchers analyze policies, procedures, and emerging technologies with the aim of improving oversight and accountability in the management of public funds (Achakzai & Peng, 2023; Ashtiani & Raahemi, 2022; Rezaee, 2005). Case studies and best practices from various jurisdictions are examined to identify effective strategies for local governments to implement, combat corruption, and promote sustainable economic development (Nasir et al., 2019).

3.8.3. **Topic 3: emerging technologies and their impact on accounting transparency**

3.8.3.1. Subtheme 3.1: NLP and sentiment analysis and accounting transparency: a new era of financial records. This subtopic examines the potential of sentiment analysis based on NLP to enhance the transparency and accuracy of financial information recording. Researchers employ advanced NLP techniques to analyze financial texts, audit reports, and press releases in search of sentiment indicators that are indicative of fraudulent practices (Hogan et al., 2008; Liu et al., 2023; Mandal & S, 2025; Perols, 2011). Sentiment

analysis and classification models are developed to identify linguistic and emotional patterns that indicate unethical behaviors or manipulations FS. Emerging technologies improve accounting transparency, the precision of financial documentation, and contribute to the detection and prevention of FSF (Bader et al., 2024; de Oliveira Orth et al., 2024; Jan, 2021; Lokanan & Sharma, 2024; Mandal & S, 2025; Nasir et al., 2019).

3.9. Emerging trends: an approach from cluster and content analysis

The simultaneous approach of cluster analysis and content analysis represents a methodological strategy that integrates the consolidated thematic structure of scientific literature with the identification of emerging trends and transforming theoretical approaches. While clusters, from a quantitative logic, reveal the centrality and connection of key terms in academic production, qualitative content analysis provides investigative narratives, applied methodologies, and specific contexts addressed by studies; this identifies gaps and future projections in research on the topic.

A keyword co-occurrence network analysis revealed nine distinct clusters, each representing a specific approach in the application of ML techniques (XGBoost, Random Forest), deep learning (CNN, LSTM), and platforms such as GLAI, IBM Watson, and DataRobot to detect fraud in SF; these approaches are complemented by expert systems and genetic algorithms to enhance accuracy and effectiveness in the analysis (Hajek & Henriques, 2017; Kootanaee et al., 2021; Nigro, 2025; Whiting et al., 2012). These clusters, visualized through a strategic diagram (Fig. 7), illustrate the diverse and interconnected landscape of FSF. The strategic diagram classifies the clusters based on centrality and density metrics, providing valuable insights into their degree of development and potential impact on accounting research and professional practice (see Fig. 8).

A fundamental aspect of this analysis is the use of the strategic diagram to represent the applications of the FSF. This visualizes the centrality and density metrics within keyword clusters in a two-dimensional graph, as shown in Table 5. Centrality reflects the sum of the correlation frequencies or the strength of the links between the keywords in a cluster and those located in other areas of the co-occurrence map, and density measures the strength of the internal connections within each cluster, which reveals the level of conceptual development of each research line (Batz et al., 2025; Callon et al., 1991).

First, the findings of topic 1 of the content analysis focused on advanced techniques for detecting FSF directly correspond to clusters 4, 5, and 6, which group studies on data mining, algorithm-based detection, and predictive risk assessment (Gupta & Mehta, 2021; Perols, 2011; Shahana et al., 2023); Kootanaee et al. (2021) and Lokanan et al. (2019) apply data mining and hybrid models based on algorithms such as SVM and ID3 for fraud detection and predictive assessment of financial risks; a parallelism that reveals an orientation toward computational and statistical models that automate the identification of anomalies in FS, gradually displacing the paradigm based exclusively on the professional judgment of the auditor (Brasel et al., 2016; Encalada, 2023; Lisic et al., 2015). Additionally, the need for ensemble learning approaches, the use of specialized datasets, and the development of hierarchical trees to classify types of fraud is highlighted, suggesting the use of analytical tools applied in forensic auditing (Jaswadi et al., 2024; López-Rojas, 2017; Mpofu, 2023). Therefore, a substantive integration between technological advancement and the need for more sensitive and effective methodological responses to the complexity of accounting fraud is evident.

Additionally, topic 2 regarding the impact of economic crises and specific sectors on the manipulation of SF is linked to clusters 1, 2, and 7, which address structural factors of corporate governance, auditor independence, and ethical conditions in accounting practices; a relationship that highlights technical elements and contextual determinants (economic, sectoral, and regulatory), significantly influencing the propensity for fraud (Kaawaase et al., 2021; Reurink, 2018; Zajmi, 2019). In this sense, the fraud triangle theory,

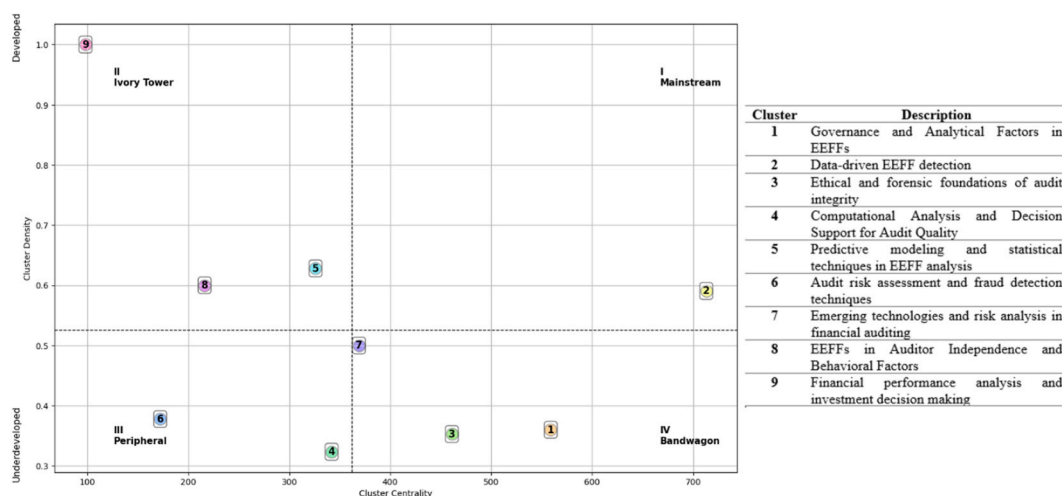


Fig. 7. Strategic Diagram Analysis

Note: Quadrant I: Mainstream (central and developed); Quadrant II: Ivory Tower (developed but peripheral); Quadrant III: Peripheral (marginal topics); Quadrant IV: Bandwagon (emerging and popular). **Source:** Authors.

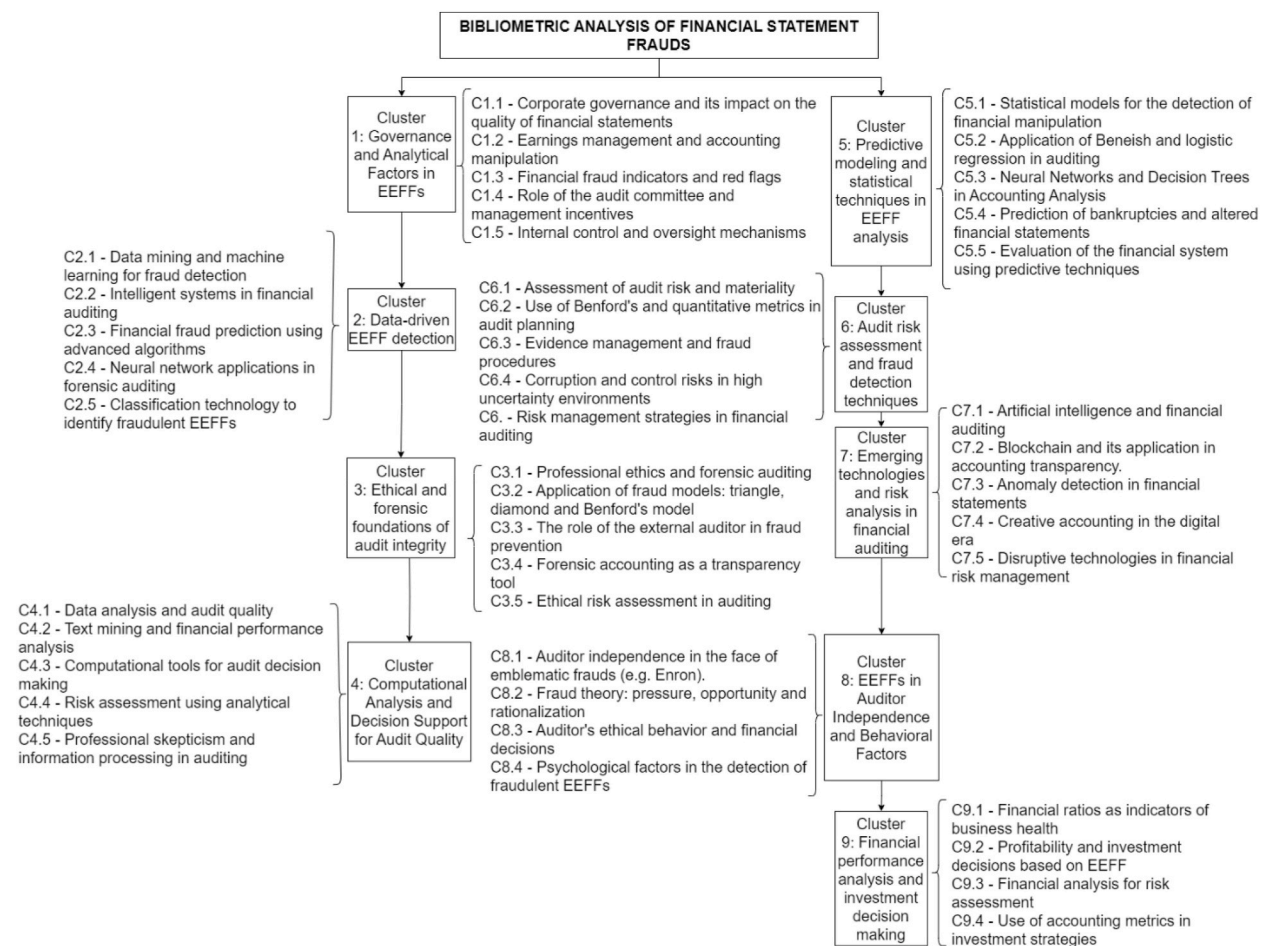


Fig. 8. Taxonomy in the bibliometric analysis of fraud in FSF.

Source: Authors

Table 5

Cluster metrics in fraud applications in FSF.

Cluster	Nodes	Edges	Centrality	Density
C1: Governance and analytical factors in FS	1066	117	559	0.3600
C2: Data-driven detection of FSF	578	124	713	0.5905
C3: Ethical and forensic foundations of audit integrity	20	54	461	0.3529
C4: Computational analysis and decision support for audit quality	644	44	342	0.3235
C5: Predictive modeling and statistical techniques in FS analysis	1025	49	326	0.6282
C6: Audit risk assessment and fraud detection techniques	1611	17	172	0.3778
C7: Emerging technologies and risk analysis in financial auditing	1020	14	369	0.5000
C8: FSFs in auditor independence and behavioral factors	1005	9	216	0.6000
C9: Financial performance analysis and investment decision-making	975	3	98	1.0000

Source: Authors

articulated around opportunity, pressure, and rationalization, gains relevance when revisited in content analyses and clusters associated with behavioral factors (Hudayati et al., 2022; Narsa et al., 2023; Parlindungan et al., 2017). The above reinforces the idea that regulatory reforms, internal control systems, and organizational culture are peripheral elements with critical dimensions in building reliable financial information systems resistant to FSF (Guevara-Sanabria et al., 2019; Leal et al., 2020; Nasir et al., 2019).

Similarly, this logic is expanded in theme 3, which addresses the impact of emerging technologies such as NLP, blockchain, and AI on accounting transparency. Although clusters 3, 8, and 9 do not focus on emerging technologies, they are indirectly/complementarily related. That is, cluster 3, focused on fraud detection and prevention, can benefit from AI applications and advanced analytics (XGBoost, Random Forest, and text mining with NLP); cluster 8, which examines human and ethical factors, can integrate with technological solutions to strengthen controls (IBM Watson for automated auditing and AI-based control systems); and cluster 9, which

analyzes financial indicators, could link with data analysis techniques to improve interpretation and accuracy, such as predictive models and supervised and unsupervised ML (Hilal et al., 2022; Maqbool et al., 2023; Nigro, 2025; Rodríguez, 2023).

On the other hand, an epistemic shift is observed here in the accounting discipline where technology serves as an instrumental complement and redefines the operational and ethical standards of auditing by providing greater traceability, automation, and real-time control (DeZoort & Harrison, 2018; Mpofu, 2023; Rashid et al., 2023). In this context, sentiment analysis applied to financial reports and the tokenization of accounting records represent rapidly expanding research lines, indicating a shift towards continuous, intelligent, and decentralized audit models (Ashfaq et al., 2022; Gaikwad et al., 2023; Qader & Cek, 2024).

Therefore, this technological transformation is functionally articulated with cluster 9 focused on profitability, financial ratios, and investment decisions; the incorporation of predictive models and automated analyses explored in previous clusters finds its practical validation in this space, by improving the quality of information available for strategic decision-making (Lokanan & Sharma, 2024; Omar et al., 2017; Wadhwa et al., 2020). Thus, the ability to generate sustainable economic value increasingly depends on the robustness of audit processes and the reliability of SF, strengthening the link between technical analysis and value creation; a cross-cutting aspect that represents the need for interdisciplinary approaches integrating finance, ethics, technology, and data analysis (Ali, Nipu, & Khan, 2023; Estep et al., 2024; Zhao & Bai, 2022).

Similarly, the analysis of the thematic clusters organized in the strategic diagram provides a structured understanding of the current state of FSF research. The grouping of themes allowed the identification of three main approaches: (1) the application of computational and statistical tools to improve audit quality (clusters 4, 5, and 6); (2) fraud detection and risk analysis in SF driven by emerging technologies and data-based techniques (clusters 1, 2, and 7); and (3) a more normative and contextual aspect encompassing professional ethics, auditor behavior in their audit engagements, and investment decisions (clusters 3, 8, and 9).

The distribution of these clusters in the quadrants of the strategic diagram reveals implications in the literature. Cluster 2 located in Quadrant I (Mainstream) indicates that the use of data to detect fraudulent SF represents a central and consolidated axis in the field. On the other hand, clusters 5, 8, and 9 in Quadrant II (Ivory Tower) show that, although statistical modeling approaches, financial performance analysis, and auditor behavioral factors have been developed, they are not yet fully integrated into the thematic core, suggesting opportunities for greater theoretical articulation. In Quadrant III (Peripheral), clusters 4 and 6 reflect areas with lower visibility, such as computational analysis and risk assessment, which could benefit from FSF studies. Finally, Quadrant IV (Bandwagon) with clusters 1, 3, and 7 highlights emerging and rapidly expanding lines of research, particularly around governance, forensic ethics, and the use of innovative technologies in auditing.

The analysis of thematic clusters and emerging topics highlights the consolidation of structured fields around financial auditing, FSF detection, and business performance evaluation, as well as the emergence of new research lines incorporating behavioral perspectives, intelligent algorithms, and ethical considerations adapted to digital contexts. Therefore, the convergence between cluster analysis and content analysis aids future research that integrates quantitative, qualitative, and computational approaches, thereby contributing to an evolution oriented towards building more transparent, efficient, and ethically robust accounting systems (see Fig. 8).

4. Discussion and implications

From an accounting and financial perspective, the bibliometric and systematic review results of this study allow for the identification of relevant implications for different stakeholder groups. The temporal patterns observed in the literature, characterized by increases in academic production following major corporate scandals and financial crises, reflect that research on accounting fraud responds to large-scale economic and social consequences. In this sense, the FSF constitutes a recurring reporting problem and, in turn, a structural factor that affects the functioning of financial markets and social welfare.

For example, the Enron bankruptcy involved the layoff of over 20,000 employees, the loss of hundreds of millions of dollars in pension savings, and the disappearance of Arthur Andersen, one of the leading global auditing firms (U.S. Senate, 2002; U.S. Securities and Exchange Commission, 2004). This case significantly eroded trust in financial information and the audit function. Similarly, the global mortgage crisis, underpinned by deceptive financial practices and systemic control failures, generated severe economic losses in numerous countries (Boubaker et al., 2016). Additionally, it affected institutional investors managing pension funds and triggered a profound crisis of confidence in the international financial system (Johnson & Peterson, 2014; Lee et al., 2014).

For companies and their management teams, the bibliometric results and content analysis of this study show that FSF episodes generate significant economic, organizational, and governance consequences. The most influential literature documents that regulatory authorities impose sanctions and that fraud disclosure causes reputational losses. A deterioration in the credibility of financial information is also observed. These consequences translate into accounting restatements, changes in reporting policies, and adjustments to internal control systems, as can be seen in clusters 1 and 2.

From the perspective of managers and senior executives, the evidence reviewed points to significant legal and professional consequences. These include dismissals, financial penalties, and increased exposure to litigation. In addition, there has been an intensification of internal and external oversight. The thematic patterns identified show that fraud scandals act as catalysts for corporate governance reforms. These reforms are reflected in the strengthening of audit committees, greater auditor independence, and increased attention to internal control mechanisms. This emphasis is recurrent in the literature following major financial crises (see clusters related to corporate governance and institutional factors, Clusters 3, 4, and 8).

For accounting professionals and auditors, the implications of this article lie in providing an empirical basis for identifying where the main risks of fraud in FS are concentrated and which areas of the accounting process require greater professional attention. The bibliometric results and content analysis show a recurring concentration of research on critical accounting principles, such as revenue

recognition, accounting estimates, and the exercise of professional judgment, along with internal control and audit weaknesses (see Clusters 1, 2, and 8). In this context, the study shows that the progressive integration of Industry 4.0 technologies, such as data analytics and artificial intelligence, is becoming established as a necessary complement to traditional auditing practices. This process favors their everyday use in fraud prevention and early detection, in line with trends identified in recent literature (see Clusters 5, 6, and 7—verify whether these are the clusters that support this part or if there are others).

For regulators and supervisory bodies, the implications translate into the need to strengthen transparency frameworks. It is essential to improve the effective enforcement of regulations and update audit and supervisory procedures. The results show that a substantial part of the investigation focuses on weaknesses in internal control, financial reporting quality, and risk assessment, areas directly linked to the effectiveness of regulatory enforcement (see Clusters 1, 2, 7, and 9). In this regard, public policies aimed at harmonizing accounting standards, strengthening supervision, and incorporating advanced analytical capabilities can help reduce incentives for fraud. Similarly, recent literature supports the design of public policies that encourage the use of tools based on data analytics and artificial intelligence to support the early detection of irregularities, in line with the trends identified in studies on detection technologies and continuous monitoring (see Clusters 5 and 6).

For academic researchers, the results of this study offer clear guidance on emerging directions and research gaps in the field of FSF. Trend analysis shows a progressive shift from predominantly descriptive approaches to more applied research focused on the design, evaluation, and implementation of specific fraud prevention and detection mechanisms (see Section 3.9). In particular, recent literature shows a growing interest in integrating accounting, financial, and technological perspectives. This approach is reflected in the increasingly frequent use of real data, advanced analytics, and artificial intelligence in the study of FSF, which reflects a transition toward research with greater practical relevance and potential impact on decision-making (see Clusters 1, 5, 6, and 7). The results highlight the need for more in-depth comparative analysis between institutional contexts, regulatory frameworks, and governance systems. It is also important to advance the study of the interactions between managerial incentives, financial reporting quality, and continuous monitoring technologies, which are emerging as central themes in the future research agenda.

5. Conclusions

Approaches were used that illustrated the study methodology to investigate the dynamics of the FSF through a bibliometric and thematic analysis, based on the literature and in relation to the conceptual, methodological, and technological transformation of the subject. Based on the findings, it should be noted that, historically, this research on FSF has been based on important findings that made fundamental contributions to establishing the principles of the conceptual foundations of fraud.

Beasley (1996), Farber (2005), and Beasley et al. (2000) determined that corporate governance, board member selection, and sector-specific characteristics are significant determinants of fraud occurrence, providing empirical support and continuing to shape contemporary research trends. Similarly, the number of publications by Brazel, Jones, and Albrecht, ranging from six to seven, documents the continuing role of an influential academic center in shaping the discipline.

Scientific output is concentrated geographically and institutionally in the United States (105 publications and high-profile institutions such as the University of Minnesota (9 articles), Arizona State University (8), and Brigham Young University (8)). However, the field shows signs of progressive diversification, with a marked increase in the participation of Asian, European, and Latin American institutions, including Nanyang Technological University (3 articles), Bina Nusantara University (8), Universiti Seni Indonesia (18), the University of Economic Studies in Bucharest (7), and Ferdowsi University in Mashhad (8). This expansion reveals a process of internationalization of financial fraud as it is studied, where different regulatory frameworks and established academic traditions intersect.

Organizational, financial, and accounting perspectives (473 articles), economics and finance (363), social sciences (228), decision sciences (92), and computer science (79) dominated most of the research articles in this thematic analysis. This distribution shows the multidisciplinary nature of the framework, which combines economic, sociological, behavioral, and technological approaches to understand and predict anomalies in financial statements.

In contrast, the high concentration of publications in specialized journals is an indicator of the concentration of the topic on the agenda of this international group. Some well-known examples are the *Journal of Financial Crime* (36 articles), the *Journal of Business Ethics* (18 articles), and the *Managerial Auditing Journal* (14 articles). In general, these contribute to the study of fraud, which is evaluated from ethical, regulatory, and computational perspectives. The overall findings are consistent with a growing literature review on the topic. The results demonstrate the increasing diversity of the field (e.g., the integration of new technologies and a variety of research areas) and the consolidated application of analytical methods, which has taken the study of financial fraud to a deeper and more preventive category.

Subsequently, the intellectual structure defined by the nine clusters above (C1–C9), together with their construction into three major emerging themes, marks a significant transformation of fraud research within FS. This development has led to the transition from a focus on governance and ethics to an analytical model largely centered on data, and subsequently to the inclusion of new Industry 4.0 technologies.

From a structural perspective, the field is based on a set of separate but interconnected thematic domains that form an evolving research ecosystem. The clusters around corporate governance, behavioral aspects, and auditor independence (C1, C3, and C8) demonstrate that human and ethical elements remain a fundamental interpretive axis through which some understanding of the conditions under which fraud occurs can be gained.

At the same time, clusters addressing quantitative analysis, statistical modeling, and risk assessment (C2, C4, C5, C6) show the establishment of a methodological institution that allows for the systematic identification of anomalies with organized data and

predictive models. Finally, clusters correlated with trends in emerging technologies, transparency, and financial decision-making (C7, C9) emerged, supporting the arrival of a new era in which automation, text mining, and algorithmic analysis are transforming auditing and the way we understand fraudulent phenomena.

The thematic synthesis of the analysis allowed this complexity to be organized into three main dimensions. In fact, the transformation of the trend around Theme 1, which addresses advanced fraud detection methods and techniques, clearly demonstrates how, in terms of methods, to analyze the disadvantages of conventional strategies using modern data, machine learning algorithms, and hybrid approaches. The sub-themes discovered, such as detecting outliers beyond the arbitrarily significant level, joint learning techniques, and the creation of special data sets and hierarchical fraud trees, indicate attempts to equip the area with more robust and replicable tools that can be used in a scenario of large volumes, large amounts of relevant information, and increasingly sophisticated fraud patterns.

The second study, which focused on the impact of crises and specific sectors on FSF manipulation, showed that fraud is not a universal or stable phenomenon, but is conditioned by the macroeconomic environment, sectoral pressures, and the regulatory system. Clusters C1, C2, and C7 indicate that in crisis situations or when sectoral vulnerabilities exist, incentives for accounting manipulation are high and risk patterns change. This contextual perspective also allows for a more flexible interpretation of fraud, as it becomes evident in the global economic context and is subject to external shocks and structural incentives that condition corporate decisions. The project's third focus, emerging technologies and their impact on accounting transparency, revealed one of the most important developments.

The use of NLP, sentiment analysis, and automated auditing of accounting narratives represents an improvement in the method applied to detect fraudulent actions in data sources that have been ignored until now, such as narrative reports, notifications in financial reports, and corporate communications. As a result, blockchain, AI, and predictive analytics promise to provide new capabilities to strengthen traceability, internal control, and the accuracy of accounting records. In this way, clusters C3, C8, and C9 show that this technological convergence is also expanding auditing practices and transforming the notion and measurement of financial transparency. The study confirms that FSF research is also growing and maturing. It effectively articulates concepts of governance, ethics, and auditing through sophisticated quantitative methodologies, establishing academic maturity. Emerging technologies are expanding to create new frontiers, requiring new concepts and flexible tools—tools that must function in complex and highly interconnected digital ecosystems.

Therefore, the study offers a more systematic compilation of intellectual advances in this field, highlights some long-standing gaps in particular, the lack of sectoral empirical studies, the lack of full adoption of explainable AI, non-standardized measures of narrative analysis results, and the lack of inclusion of behavioral factors in automation, and describes a roadmap that guides future research agendas. It also highlights the need to move toward multidisciplinary research, combining elements of accounting, data science, computational linguistics, organizational ethics, and systems engineering, to improve the quality, transparency, and integrity of financial information.

In summary, the results indicate a transition toward a more hybrid model in which quantitative, technological, and ethical aspects intersect. This study facilitates the consolidation of a conceptual and methodological framework to better determine, predict, and mitigate FSF in a globalized environment known for its growing demands for transparency and reliability with the articulation of trends, mapping of theoretical links, and summarized research emergencies.

Limitations

The research has some limitations due to the limitations of the literature search using the term FSF, so it is possible that relevant studies using different terminology or perspectives may have been overlooked. By focusing on management, business, economics, and social sciences, this study primarily analyzes fraud from an organizational and economic perspective, which means that perspectives from more technical fields such as computer science or engineering may have been overlooked. In addition, the review only considered studies indexed in Scopus and Web of Science, excluding research published in other journals or languages, which limits the variety of perspectives included. The rapid pace of technological change also means that some findings may not be relevant in the long term. Even with these limitations, the study offers an overview of the topic and provides a useful basis for future research aimed at generating greater transparency and confidence in financial information.

The advisability of evaluating the actual effectiveness of advanced data analysis tools in different organizational contexts; the lack of evidence on the impact of training in ethics and fraud detection on the skills of accountants and auditors; and the need to further investigate the effect of emerging technologies, such as continuous auditing, digital traceability, and blockchain, on the transparency and reliability of financial information.

Ethics approval and consent to participate

Not applicable.

Consent for publication

All participants consented to the publication of anonymized responses and findings derived from this study.

Availability of data and material

The data that support the findings of this study are openly available in the Harvard Dataverse repository at <https://doi.org/10.7910/DVN/VQC8KW>.

Authors' contributions

All authors read and approved of the final manuscript.

Declaration of use of generative AI

During the preparation of this work, the author(s) used RUBRIQ™, part of Springer Nature, in order to improve the English language, grammar, punctuation, spelling, and overall style. After using this service, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Competing interests

The authors declare that they have no competing interests.

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Data availability

Data will be made available on request.

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